BSCS 5th Semester | PUACP | Web Technologies – EC-331
Topic: Web Services: Operations, Processing HTTP Requests, Processing HTTP Responses, Cookie Coordination, Privacy and P3P, Complex HTTP Interactions.
By Noorayy Jam

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1. Introduction to Web Services

Web services are standardized technologies that enable different applications to communicate and share data across networks, regardless of their platforms or programming languages. They form the backbone of modern internet-based communication by allowing interaction between client applications (like browsers or mobile apps) and server-side systems through the use of common protocols such as HTTP, XML, JSON, and SOAP.

A web service operates as a bridge between two systems, ensuring smooth data exchange. For example, when a weather application fetches live weather data, it does so through a web service request to a remote server that processes the request and sends back an appropriate response.

Web services are typically divided into two major types:

1. **SOAP (Simple Object Access Protocol):** A protocol that uses XML-based structured messaging and focuses on security, reliability, and strict communication standards.
2. **REST (Representational State Transfer):** An architectural style that uses simple, lightweight, and stateless communication over HTTP using methods like GET, POST, PUT, and DELETE.

Both play essential roles in system integration, allowing different software components to work cohesively within enterprise systems, e-commerce platforms, and cloud applications.

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2. Web Service Operations

The operations of a web service describe the complete lifecycle of communication between a client and a server. These operations ensure that requests are handled properly, data is processed accurately, and appropriate responses are delivered back to the client. The main steps include:

1. **Request Initiation:** The client (browser or app) sends a request to a server through an HTTP or HTTPS protocol. This request specifies what resource or operation is desired, such as retrieving data or submitting a form.
2. **DNS Resolution:** Before the server can be reached, the domain name (e.g., www.puacp.edu.pk) is resolved into its IP address through the Domain Name System (DNS).
3. **TCP Connection Establishment:** The client and server create a communication channel through the TCP three-way handshake.
4. **Request Transmission:** The client sends an HTTP request, including a request line, headers, and an optional body containing data.
5. **Server-Side Processing:** The server interprets the request, executes the required logic using backend languages such as PHP, Python, or Node.js, and communicates with the database if needed.

6. **Response Generation and Delivery:** The processed data is packaged into an HTTP response and sent back to the client.
7. **Rendering:** The client's browser or application processes the response and displays it to the user.

Each operation plays a vital role in ensuring data reliability, accuracy, and security. Web service operations must adhere to strict communication protocols to maintain interoperability between different systems.

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3. Processing HTTP Requests

An **HTTP Request** is a message sent by a client to the server to request access to specific resources such as web pages, images, or data. It is a structured message that follows a specific format and consists of three main components: the request line, headers, and an optional body.

Structure of an HTTP Request:

```
GET /index.html HTTP/1.1
Host: www.puacp.edu.pk
User-Agent: Chrome/118.0
Accept: text/html
Authorization: Bearer token123
Connection: keep-alive
```

Components of an HTTP Request:

- **Request Line:** This line includes the HTTP method (GET, POST, PUT, DELETE), the resource path, and the HTTP version. It defines the specific operation the client wants to perform.
- **Headers:** Headers provide additional information to the server about the client environment, content type, accepted formats, and authentication details.
- **Body (Optional):** Contains the main data to be transmitted, usually included in POST or PUT requests (for example, user credentials, JSON objects, or form submissions).

Common HTTP Methods:

- **GET:** Retrieve data from the server.
- **POST:** Send new data to the server.
- **PUT:** Update an existing resource on the server.
- **DELETE:** Remove a specific resource.
- **PATCH:** Make partial updates to an existing resource.

Each HTTP method serves a specific purpose and follows defined semantics that ensure proper resource manipulation. HTTP requests form the foundation of client-server communication, enabling every form of online interaction, from form submission to data retrieval.

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4. Processing HTTP Responses

After the server receives and processes an HTTP request, it sends back a structured **HTTP Response**. This response communicates the outcome of the request and may include the requested data, metadata, or an error message.

Structure of an HTTP Response:

```
HTTP/1.1 200 OK
Content-Type: text/html
Content-Length: 1256
Set-Cookie: session_id=abc123; Secure; HttpOnly

<html>
<body>Welcome to PUACP Web Technologies!</body>
</html>
```

Components of an HTTP Response:

1. **Status Line:** Contains the HTTP version, status code, and reason phrase (e.g., 200 OK).
2. **Headers:** Provide metadata about the response, such as content type, length, and server information.
3. **Body:** The actual content, which may be an HTML document, image, JSON, or XML data.

Common HTTP Status Codes:

- **1xx (Informational):** Request received and continuing process.
- **2xx (Success):** Request was successful (e.g., 200 OK, 201 Created).
- **3xx (Redirection):** Further action required (e.g., 301 Moved Permanently, 302 Found).
- **4xx (Client Error):** Problem with client request (e.g., 400 Bad Request, 404 Not Found).
- **5xx (Server Error):** Server failed to fulfill a valid request (e.g., 500 Internal Server Error, 503 Service Unavailable).

HTTP responses ensure effective communication between the client and the server, conveying both success and failure messages with detailed diagnostic information.

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5. Cookie Coordination and Session Management

HTTP is a stateless protocol, meaning it does not retain any information about previous interactions. To overcome this limitation, **cookies** are used to maintain user sessions and preferences.

What Are Cookies?

Cookies are small text files stored in the user's browser that contain information about the user's session. They help identify returning users and maintain stateful communication between the client and server.

Types of Cookies:

1. **Session Cookies:** Temporary cookies deleted when the browser is closed.
2. **Persistent Cookies:** Remain stored on the device until their expiration date.
3. **Secure Cookies:** Transmitted only over HTTPS for enhanced security.
4. **HttpOnly Cookies:** Inaccessible to JavaScript, preventing cross-site scripting (XSS) attacks.
5. **Third-Party Cookies:** Set by external domains, often used for analytics or advertisements.

Session Management Techniques:

- **Cookies:** Store session identifiers on the client side.
- **URL Rewriting:** Appends session IDs to URLs.
- **Hidden Form Fields:** Transmit session data within forms.

Example:

```
Set-Cookie: session_id=xyz789; Expires=Wed, 09 Oct 2025 12:00:00 GMT; Path=/;
Secure; HttpOnly
```

Proper cookie coordination ensures secure, continuous, and personalized user interactions, essential for applications like banking, e-commerce, and social networks.

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6. Privacy and P3P (Platform for Privacy Preferences)

The **Platform for Privacy Preferences (P3P)** was introduced by the World Wide Web Consortium (W3C) to make website privacy policies readable and interpretable by browsers. It allowed users to understand how their personal information was collected, stored, and used.

Purpose of P3P:

- To promote transparency in data handling practices.
- To enable browsers to automatically compare user privacy preferences with website policies.

Structure of a P3P Policy:

1. **Data Collection:** Defines what personal data (like names, emails, cookies) is collected.
2. **Purpose:** States why data is being collected (e.g., analytics, marketing, authentication).
3. **Recipient:** Identifies who receives the data (first-party or third-party).
4. **Retention:** Specifies how long the data is stored.

Example Header:

```
P3P: CP="CURa ADMa DEVa TAIa OUR BUS IND PHY ONL UNI COM NAV INT DEM PRE"
```

While P3P has been deprecated in modern browsers, it remains an important historical step toward developing web privacy standards and user trust frameworks.

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7. Complex HTTP Interactions

Modern web communication involves more than simple request-response cycles. Complex HTTP interactions occur when multiple intermediaries, redirections, or authentication mechanisms come into play.

Key Examples:

- **Redirects (3xx Codes):** Servers can instruct clients to access a different resource location.
- **Proxies and Gateways:** Act as intermediaries for security, caching, or load distribution.
- **Caching Layers:** Temporarily store responses to minimize latency and reduce server load.
- **Load Balancers:** Distribute incoming requests across multiple servers for scalability.
- **Authentication Flows:** Use OAuth2, JWT tokens, or session cookies for secure access.
- **CORS (Cross-Origin Resource Sharing):** Allows controlled data sharing between different domains.

Example Header:

```
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: GET, POST, PUT
```

Complex interactions enable efficient, secure, and high-performing web systems that can handle large volumes of user requests simultaneously.
